HDOC Day-70

Question: Write a recursive function to calculate the digital root of a number. Use a static variable to count the number of recursive calls.

C Program

```
#include <stdio.h>

int digitalRoot(int n) {
    static int calls = 0;
    calls++;

    if (n < 10) {
        printf("Recursive calls: %d\n", calls - 1);
        return n;
    }

    int sum = 0;
    while (n > 0) {
        sum += n % 10;
        n /= 10;
    }

    return digitalRoot(sum);
}

int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);

    if (num < 0) {
        printf("Please enter a non-negative number.\n");
        return 0;
    }

    printf("Digital root: %d\n", digitalRoot(num));
    return 0;
}
```

Sample Output

```
Enter a number: 9875
Recursive calls: 3
Digital root: 2
```

Explanation

9875 = 9 + 8 + 7 + 5 = 29
29 = 2 + 9 = 11
11 = 1 + 1 = 2

The function repeatedly adds the digits until a single digit remains. The static variable keeps its value across function calls. The initial function call is not counted as a recursive call.